Summary of Findings

Does your suspect's alibi hold up?

Compare each of the samples taken from your suspect with the samples taken from the location where the victim was found and from the victim.

V1 the beach where Stan was found = S1B = S2C Suspect 1's car and fingernails of Suspect 2

V2 (inside skull) = S2B = S1D = S4A (beach at locality 4)

V3 (campsite) = S1C= S4B =

Compare each of the samples taken from your suspect with the samples taken from where they claimed to be.

Which could not be from the same place and why?

Suspect 1: S1A, S1E – consistent with a mountainous area – and Digimap confirms the identification of the igneous rock.

However, S1B is from the beach where Stan was found – the car had been there as it was in the tyre tread

S1C – is the campsite

S1D – someone in the car had been to the beach at location 4

Suspect 2: No evidence they were at the beach at location 2. The samples from their clothing and body indicate they were at the beach at location 4 and the beach where the body was found.

Suspect 3: S3C – is the pub car park, and S3B is not the same as the beach at location 3

Suspect 4: S4B – combination of sand from the campsite and around the pub, the plant *Limonium britannicum* is only found at location 4 so he must have been there.

As a Group decide:

Be sure to give evidence for your answers

1. Where was the victim killed?

V2 (inside skull) = S2B = S1D = S4A (beach at locality 4)

2. Who did it?

Giles – his trousers were at location 4 and his fingernails show he was at the beach where Stan's body was found.

He needed a car to transport the body from location 4 and so Dave may have been involved, his car certainly was.

3. How confident are you in your interpretation/conclusions?

The evidence from the sediment samples could certainly add to the police's case against Giles, but not 100% certain.

4. What additional data would you like to collect to aid your interpretation?

Statistical analysis of the sand samples would be more accurate – see Week 10

Sample 1C

Grain shape: Spherical

Angularity: Variable: 90% rounded-well rounded 10% subangular (tend to be grains

Composition: 95% quartz 5% calcite (occur at fine end of grain size distribution)

Other observations Grain surface is frosted, Fe oxide staining present

Sample 2B

Grain shape: Shell frags: platey Metamorphic/ignous lithics: subsphericalplatey Quartz: subspherical

Angularity v.angular-subangular

Composition 40% shell fragments 10% coralline algae 40% quartz 20% basic igneous (including gabbro) and metamorphosed basic intrusive igneous

Sample 3A

Grain shape Sub-spherical

Angularity Rounded to subrounded

Composition 100% quartz

Sample 4B

Grain shape Population 1:spherical Population 2: subspherical

Angularity Population 1: 90% rounded-well rounded 10% subangular (tend to be grains

Composition Population 1:95% quartz 5% calcite (occur at fine end of grain size distribution) Grain surfaces are frosted. Fe oxide staining present Population 2: 100% quartz (each clast is an aggregates of quartz grains)

Bimodal grain size population!!